# GraVLM: A Hierarchical Graph-Aligned Vision–Language Model for Cross-Modal Retrieval in Remote Sensing

Xi Chen, Xu Chen, Wei Wang, Xiangyang Jia, Gongping Yu, and Feng Li, *Member, IEEE*

***Abstract*—Cross-modal text–image retrieval in remote sensing (RS) aims to align RS images with natural-language descriptions for efficient access to large-scale RS image archives. Benefiting from large-scale image–text pretraining, recent vision–language models (VLMs) have demonstrated strong capability in modeling global scene-level semantics and have become a dominant paradigm for RS cross-modal text–image retrieval. However, unlike natural images, a single-RS image often contains multiple objects organized in complex spatial layouts, whereas existing RS-oriented retrieval VLMs mainly emphasize scene-level alignment and lack explicit modeling of object-level entities and their spatial relations. To overcome these limitations, we present GraVLM, a hierarchical graph-aligned VLM for fine-grained cross-modal retrieval in RS. GraVLM adopts a hierarchical design that integrates scene-level cross-modal interaction and object-level structural alignment. At the scene-level, we introduce a cross-modal interaction mechanism (CMIM) to capture the global semantic correspondence between visual and textual modalities. At the object-level, a spatial object graph (SOG) and an entity relation graph (ERG) are constructed from the visual and textual modalities, respectively, enabling graph-aligned modeling of object-centric spatial relations. Furthermore, we introduce a local-guided feature fusion (LGFF) module that performs local–global feature fusion over the SOG and the ERG to obtain cross-modal fusion embeddings. Based on these fused representations, we propose a graph-aligned contrastive loss that jointly enforces object–entity correspondence and spatial–relation consistency between the two graphs. Extensive experiments demonstrate that GraVLM consistently and significantly outperforms existing RS cross-modal retrieval models.**

***Index Terms*—Cross-modal retrieval, graph-based alignment, hierarchical modeling, remote sensing (RS), vision–language models.**

## I. Introduction

THE accelerated development of Earth's observation technologies has led to the explosive growth of remote sensing (RS) images, which play an increasingly important role in large-scale geospatial analysis and intelligent decision-making [1], [2], [3]. With the rapid accumulation of RS images, natural language-driven retrieval of relevant visual content has become a core requirement for practical RS analytics systems. As a result, cross-modal text–image retrieval has emerged as a fundamental task for bridging RS images and textual descriptions, allowing users to search, analyze, and interpret large-scale RS image archives in a more intuitive manner.

Benefiting from large-scale image–text pretraining, vision–language models (VLMs) learn global scene-level representations and achieve effective cross-modal alignment [4], [5], making them a natural choice for RS cross-modal text–image retrieval. By jointly embedding RS images and textual descriptions into a shared semantic space, VLM-based methods support flexible matching between visual content and natural-language queries without relying on task-specific handcrafted features. In particular, current VLM-based methods are effective at capturing global scene-level semantics, which already delivers strong retrieval performance at the holistic level. Recent studies further indicate that adapting general-purpose VLMs or pretraining domain-specific RS-VLMs on large-scale RS image–text datasets can substantially improve retrieval accuracy and generalization across diverse RS scenarios [4], [6]. Meanwhile, VLMs have recently been explored as transferable multimodal priors for label-efficient RS downstream tasks (e.g., few-shot segmentation and SAR automatic target recognition), demonstrating improved the generalization under limited supervision and cross-modality domain shifts [7], [8].

Despite the strong progress achieved by VLM-based methods, effectively aligning RS images and textual descriptions remains inherently challenging due to the unique structural characteristics of RS scenes. Unlike natural images that are often dominated by a small number of salient foreground objects [6], a single-RS image often contains multiple objects organized in complex spatial layouts with significant scale variations [60]. Correspondingly, RS textual descriptions tend to emphasize object-level entities and their spatial relations, such as relative positions, adjacency, and functional layouts, rather than purely scene-level features [9], [10]. However, most existing RS-oriented VLM retrieval methods predominantly focus on global scene-level alignment [4], as illustrated in Fig. 1(a), without explicitly modeling object-level entities and

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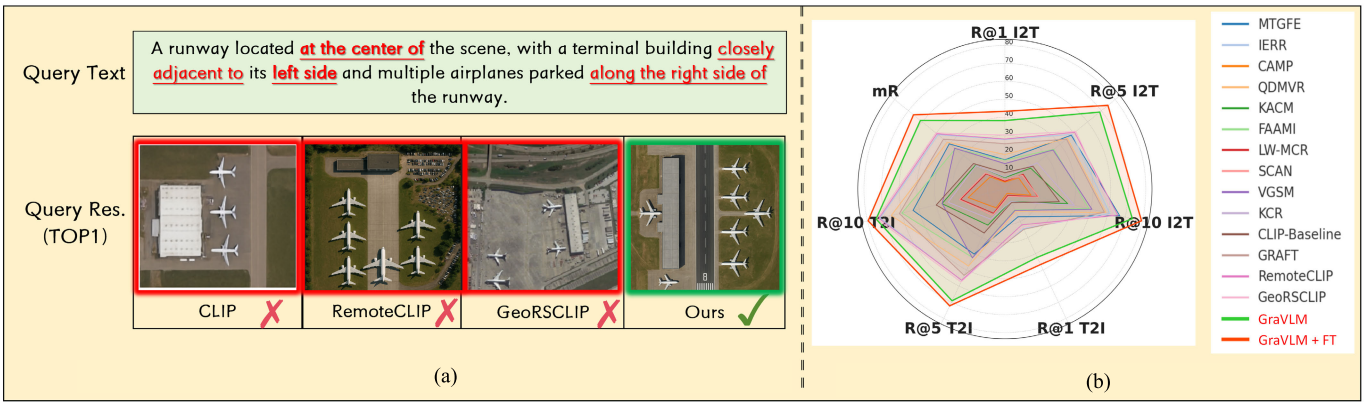



Fig. 1. (a) Comparison of spatial relations T2I retrieval results across different methods, indicating that existing models struggle to align entity-level semantics with spatial relations. (b) Performance comparison of GraVLM with state-of-the-art methods.

interobject spatial relations. As a result, these methods often struggle to capture fine-grained object correspondence and relational consistency across modalities, which are essential for resolving ambiguities in complex RS scenes and achieving the accurate cross-modal retrieval.

To address these limitations, we propose GraVLM, a hierarchical graph-aligned VLM for fine-grained cross-modal retrieval in RS. GraVLM adopts a hierarchical design that jointly integrates scene-level cross-modal interaction and object-level structural alignment within a unified model. At the scene-level, GraVLM introduces a cross-modal interaction mechanism (CMIM) to model global semantic correspondence between RS images and textual descriptions. At the object-level, it constructs a spatial object graph (SOG) from visual inputs and an entity relation graph (ERG) from textual descriptions, respectively, enabling explicit graph-aligned modeling of object-centric spatial relations. To further enhance the feature integration, GraVLM incorporates a local-guided feature fusion (LGFF) module, which fuses scene-level and object-level features, ensuring a comprehensive alignment of both global and local representations. Finally, a hierarchical graph-aligned contrastive loss (HGAC-Loss) is applied to jointly optimize object–entity correspondences and spatial relations across the two modalities. By aligning scene-level semantics and object-level graph structures, GraVLM establishes fine-grained cross-modal correspondence beyond holistic scene representations, leading to more accurate and interpretable RS cross-modal retrieval, as illustrated in Fig. 1(b).

Our main contributions are summarized as follows.

1) We propose GraVLM, a hierarchical graph-aligned VLM for RS cross-modal retrieval. GraVLM integrates scene-level cross-modal interaction with object-level structural modeling, enabling fine-grained alignment between RS images and textual descriptions beyond global scene semantics.
2) We introduce graph-based object-level representations for cross-modal, including SOG for visual images and ERG for textual descriptions. These graphs encode object-centric spatial relations, providing a unified structural foundation for cross-modal alignment in RS.
3) We propose LGFF, the key idea is to realize complementary scene–object representations in a shared embedding space: scene-level context supplies semantic background for local objects, while object-level evidence encourages the model to focus on salient objects within scenes.
4) We design a graph-aligned contrastive loss HGAC. This loss integrates node semantic supervision with relation-level structural constraints, encouraging the learned representations to preserve both fine-grained entity semantics and their spatial organization, thereby improving discriminability in RS cross-modal retrieval.

The remainder of this article is organized as follows. Section II reviews related work on RS cross-modal retrieval. Section III provides a detailed description of the proposed GraVLM. Section IV reports extensive experimental results with in-depth analyses. Finally, Section V concludes this article and outlines directions for future work.

## II. Related Works

RS cross-modal retrieval aims to bridge the semantic gap between RS images and textual descriptions. From a modeling perspective, existing methods can be broadly grouped according to whether they rely on VLMs.

### A. Methods Without VLMs

These methods are typically trained directly on RS image–text datasets using supervised or contrastive learning, without leveraging general-purpose VLMs. They can be further divided into the following two categories.

1) Global scene-level models focus on learning modality-specific global representations for RS images and textual descriptions, which are then aligned through contrastive objectives. Representative methods include AMFMN [11], KACM [12], FAAMI [13], HAC [14], MSCMAT [15], and SIPN [16]. Although these methods are effective in capturing coarse scene-level semantics and global contextual cues, they typically rely on holistic representations and lack explicit object-level modeling, which may lead to dispersed feature distributions and suboptimal retrieval performance in complex scenes.

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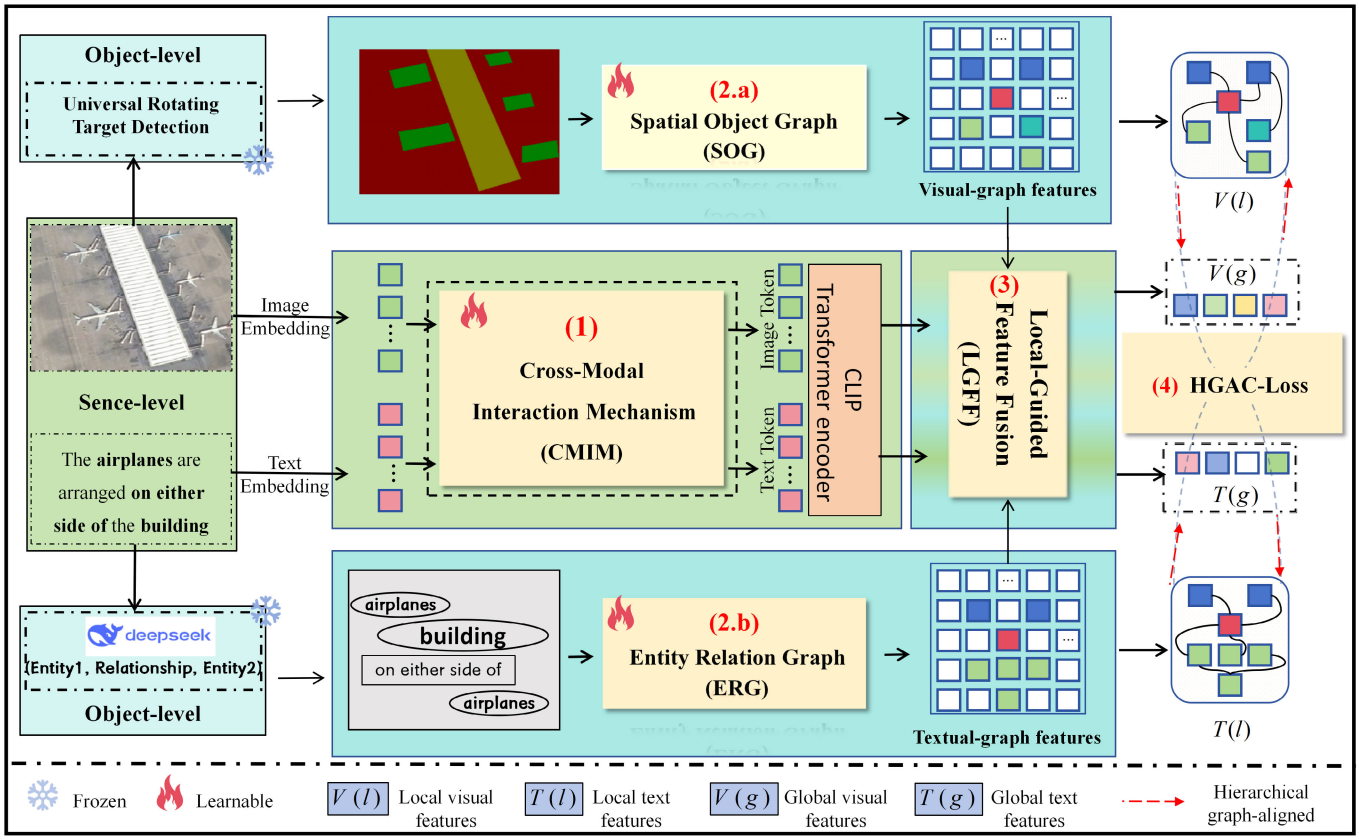



Fig. 2. Overview of the proposed GraVLM for RS cross-modal retrieval: 1) CMIM for scene-level semantic correspondence; 2) SOG (2.a) and ERG (2.b) for object-level structural modeling; 3) LGFF for local–global cross-modal embedding; and 4) HGAC-loss for aligning object–entity correspondences and object-centric spatial relations.

2) Local scene-level models extend global methods by incorporating object-level cues or localized visual reasoning. Representative works include GaLR [17], MTGFE [18], VGSM [19], SPCA-Net [20], SLAMS [21], and FSSN [22]. While these methods improve local visual representation, they often model objects independently and rarely consider interobject spatial relations or cross-modal structural consistency, which limits their ability to support relational and compositional retrieval in complex RS scenes.

### B. Methods With VLMs

Recent advances in large-scale vision–language pretraining have introduced a new category of RS cross-modal models that adapt pretrained VLMs to the RS domain. Representative methods such as RemoteCLIP [4], EarthVQA [5], and GeoChat [23] leverage large image–text corpora for retrieval, captioning, and VQA tasks [24], [25], [26]. Despite their strong generalization capability, most existing RS VLMs still emphasize global scene-level alignment and lack explicit modeling of object-level spatial relations, interobject dependencies, and compositional semantics, which are crucial for fine-grained RS cross-modal retrieval.

In summary, although prior studies have made progress in RS cross-modal text–image retrieval, three key challenges remain unresolved: 1) insufficient object-level semantic grounding; 2) limited modeling of spatial and relational dependencies among objects; and 3) absence of object-level cross-modal alignment mechanisms.

## III. Methodology

To address the above challenges, we propose GraVLM, a hierarchical graph-aligned VLM. As shown in Fig. 2, GraVLM is designed as a dual-branch architecture that simultaneously captures scene-level and object-level semantics. To integrate these complementary representations and enforce consistency across modalities, we introduce HGAC-Loss that jointly optimizes global and local structural matching.

### A. Global Scene-Level Modeling

To capture scene-level semantic correlations between visual and textual modalities, we adopt a CMIM, as illustrated in Fig. 3, which alternates modality-specific self-attention to enhance the global representation alignment.

Given an input image $I$ and its corresponding textual description $T$, we extract initial embeddings $v_0 \in \mathbb{R}^{N_v \times d}$ and $t_0 \in \mathbb{R}^{N_t \times d}$ using pretrained vision and language encoders, where $N_v$ and $N_t$ denote the number of visual and textual tokens, and $d$ is the feature dimension.

To model intramodal dependencies, we apply multihead self-attention followed by feed-forward projection:

$$\hat{v} = \text{V-FFN}\,(\text{MHA}\,(v_0))$$
$$\hat{t} = \text{L-FFN}\,(\text{MHA}\,(t_0)) \quad (1)$$

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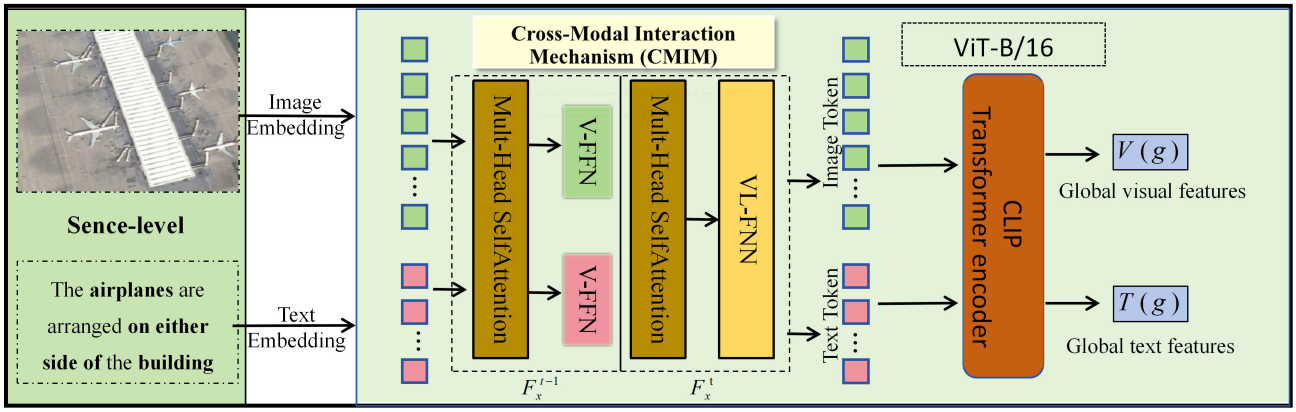



Fig. 3. Illustration of the proposed CMIM for global scene-level modeling. Intramodal self-attention captures contextual dependencies within each modality, while the cross-modal fusion enables information exchange between visual and textual tokens, resulting in globally aligned scene-level representations.

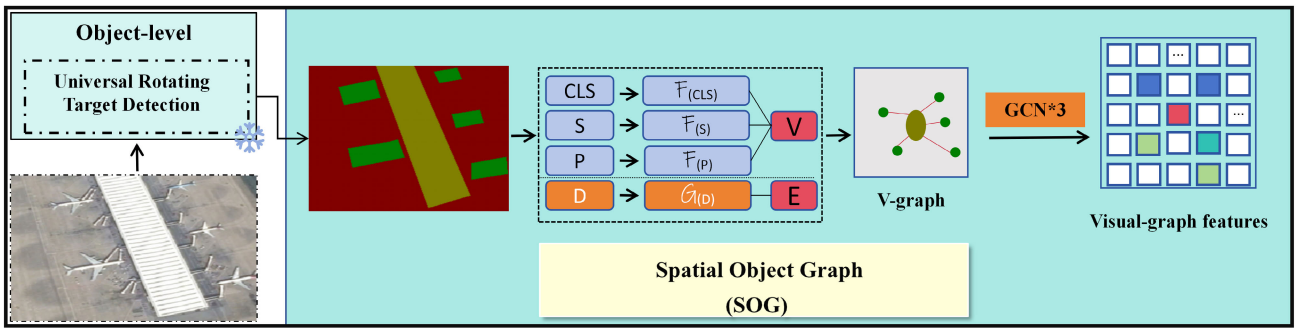



Fig. 4. Architecture of the SOG module. Each object is encoded into node features based on category, geometry, and position. Edges are constructed using inverse spatial distance. A three-layer GCN propagates local context and enhances object-level visual semantics.

where MHA(·) denotes the multihead self-attention, and V-FFN and L-FFN are the two-layer feed-forward networks with ReLU activation for each modality.

To enable the intermodal interaction, we adopt a shared cross-modal fusion block

$$f_{\text{joint}} = \text{VL-FFN}\left(\text{Concat}\left(\hat{v}, \hat{t}\right)\right) \tag{2}$$

where Concat(·) denotes the tokenwise concatenation across modalities, and VL-FFN is a shared feed-forward network for cross-modal fusion. The output $f_{\text{joint}} \in \mathbb{R}^{(N_v+N_t)\times d}$ encodes the fused joint representation.

This alternating process is repeated for $L$ layers to refine both visual and textual features. The final scene-level global embeddings are obtained as follows:

$$\begin{aligned} V^{(g)} &= \text{Agg}_v\left(f_{\text{joint}}\right) \\ T^{(g)} &= \text{Agg}_t\left(f_{\text{joint}}\right) \end{aligned} \tag{3}$$

where $\text{Agg}_v(\cdot)$ and $\text{Agg}_t(\cdot)$ denote the modality-specific aggregation functions (e.g., mean pooling over visual or textual tokens), yielding $V^{(g)}, T^{(g)} \in \mathbb{R}^d$. These global embeddings serve as the scene-level representation for subsequent coarse-grained alignment and contribute to the final contrastive loss.

### B. Local Object-Level Modeling

Motivated by the need to model fine-grained object structures and cross-modal alignment, we design a two-part object-level modeling: SOG for visual spatial reasoning, ERG for textual semantic composition.

*1) Spatial Object Graph:* To model the fine-grained spatial compositionality of RS scenes, we construct SOG for each image (as illustrated in Fig. 4). Unlike conventional methods that treat object features as independent elements, our SOG explicitly captures geometric configurations and spatial interactions between objects, which are crucial for resolving dense arrangements and directional cues in RS images.

Let the input image be $I \in \mathbb{R}^{H\times W\times 3}$, where $H$ and $W$ are the height and width, respectivley. We apply a rotated object detection model [27] to extract a set of $m$ oriented objects

$$\mathcal{O} = \{o_i = (c_i, p_i, x_i, y_i, w_i, h_i, \theta_i)\}_{i=1}^{m}. \tag{4}$$

Each object is characterized by its category $c_i$, confidence score $p_i$, center location $(x_i, y_i)$, size $(w_i, h_i)$, and orientation $\theta_i$ (clockwise from horizontal). For spatial modeling, we project all coordinates into a normalized spatial plane $[0, 1]^2$ using affine scaling

$$\left[x_i', y_i'\right] = \left[\frac{x_i}{W}, \frac{y_i}{H}\right], \quad \text{for all } i = 1, \ldots, m. \tag{5}$$

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The input image $I$ is also transformed into a patchwise grid $\mathbf{I}_g \in \mathbb{R}^{P\times P\times 3}$, where each patch corresponds to a localized semantic region

$$\mathbf{I}_g = \begin{bmatrix} I_{1,1} & I_{1,2} & \cdots & I_{1,P} \\ I_{2,1} & I_{2,2} & \cdots & I_{2,P} \\ \vdots & \vdots & \ddots & \vdots \\ I_{P,1} & I_{P,2} & \cdots & I_{P,P} \end{bmatrix} \in \mathbb{R}^{P^2\times 3} \tag{6}$$

where $I_{i,j} \in \mathbb{R}^3$ is the RGB vector of the $(i, j)$th patch.

Each object $o_i$ is encoded as a node $v_i \in \mathbb{R}^d$ by concatenating categorical and spatial descriptors

$$v_i = \mathbf{W}_f \cdot \left[\mathbf{F}_{\text{cls}}(c_i) \,\|\, \mathbf{F}_{\text{shape}}(w_i, h_i, \theta_i) \,\|\, \mathbf{F}_{\text{pos}}\left(x_i', y_i'\right)\right]. \tag{7}$$

Here, $\mathbf{W}_f \in \mathbb{R}^{d\times d'}$ is a learnable projection. Specifically, each object is represented by three components:

1) $\mathbf{F}_{\text{cls}}(c_i) \in \mathbb{R}^{d_1}$: One-hot encoding for object category.
2) $\mathbf{F}_{\text{shape}} \in \mathbb{R}^{d_2}$: Encodes object size and orientation as $[w_i, h_i, \sin\theta_i, \cos\theta_i]$.
3) $\mathbf{F}_{\text{pos}} \in \mathbb{R}^{d_3}$: Normalized position vector $[x_i', y_i']$.

The resulting object feature matrix is

$$X = \left[v_1^\top, v_2^\top, \ldots, v_m^\top\right]^\top \in \mathbb{R}^{m\times d}. \tag{8}$$

To capture spatial dependencies, we define the adjacency matrix $A \in \mathbb{R}^{m\times m}$ based on inverse squared distances between object centroids

$$A_{ij} = \frac{1}{1 + \left\|P_i - P_j\right\|_2^2}, \quad P_i = \left(x_i', y_i'\right). \tag{9}$$

This formulation prioritizes local interactions while suppressing long-range noise. We then perform symmetric normalization for stability

$$\hat{A} = D^{-1/2} A D^{-1/2}, \quad D_{ii} = \sum_{j=1}^{m} A_{ij}. \tag{10}$$

We apply a three-layer graph convolutional network (GCN) to propagate information across nodes

$$H^{(k+1)} = \sigma\left(\hat{A} H^{(k)} W^{(k)}\right), \quad H^{(0)} = X \tag{11}$$

where $W^{(k)}$ is the layer-specific weights and $\sigma(\cdot)$ is ReLU. The output feature matrix after $L = 3$ layers is

$$V^{(l)} = H^{(3)} \in \mathbb{R}^{m\times d}. \tag{12}$$

Unlike natural image scenes with sparse foregrounds, RS images often exhibit dense, overlapping, and rotationally varied object distributions (e.g., aircraft and vehicles). Our SOG design accommodates these characteristics through orientation-aware feature encoding and distance-decayed graph connectivity. This enables localized semantic reasoning that complements coarse global embeddings and facilitates fine-grained cross-modal alignment in subsequent stages.

*2) Entity Relation Graph:* To capture the compositional semantics and relational structure of RS descriptions, we introduce ERG, which transforms unstructured textual annotations into structured graph representations. As illustrated in Fig. 5, this process enables fine-grained reasoning over entity semantics and their contextual relations, providing a strong foundation for object-level cross-modal alignment.

Given an input text $T$ describing the RS scene, we employ a prompt-driven language model (DeepSeek) to extract relation triplets in the form

$$\mathcal{T} = \left\{\left(e_i^{(1)}, r_i, e_i^{(2)}\right)\right\}_{i=1}^{n} \tag{13}$$

where $e_i^{(1)}$ and $e_i^{(2)}$ are the subject and object entities, and $r_i$ is their relationship (e.g., "on either side of"). All textual spans are constrained to originate from the input description to ensure faithfulness. These triplets form the structural basis of the textual graph.

Each entity is embedded using a frozen CLIP text encoder, while relations are encoded via a dedicated linear projection

$$\begin{aligned} v_{e_i} &= \text{CLIP}_{\text{text}}(e_i) \in \mathbb{R}^d \\ v_{r_i} &= \mathbf{W}_r \cdot \phi(r_i) \in \mathbb{R}^d \end{aligned} \tag{14}$$

where $\phi(\cdot)$ is a learnable embedding for relation tokens and $\mathbf{W}_r \in \mathbb{R}^{d\times d_r}$ is a transformation matrix. The entity set yields a node matrix

$$X = \left[v_{e_1}^\top, v_{e_2}^\top, \ldots, v_{e_n}^\top\right]^\top \in \mathbb{R}^{n\times d}. \tag{15}$$

We define a textual graph $G_t = (V, E)$, where $V$ contains the extracted entities, and $E$ denotes semantic relations. The adjacency matrix $A \in \mathbb{R}^{n\times n}$ is defined via relation similarity

$$A_{ij} = \cos\left(E_{ij}, E_{ji}\right) = \cos\left(v_{r_{ij}}, v_{r_{ji}}\right). \tag{16}$$

To ensure compatibility with vision graph convolution, we apply symmetric normalization

$$\hat{A} = D^{-1/2} A D^{-1/2}, \quad D_{ii} = \sum_{j=1}^{n} A_{ij}. \tag{17}$$

The semantic structure of the ERG is encoded by GCNs

$$H^{(k+1)} = \delta\left(\hat{A} H^{(k)} W^{(k)}\right), \quad H^{(0)} = X \tag{18}$$

where $W^{(k)} \in \mathbb{R}^{d\times d}$ is the learnable weight matrix and $\delta(\cdot)$ is the ReLU activation. The final entity features are

$$T^{(l)} = H^{(3)} \in \mathbb{R}^{n\times d}. \tag{19}$$

Unlike general NLP settings, RS descriptions emphasize spatial structure and scene layout (e.g., "airplanes on either side of the building"). Therefore, we prioritize extracting spatial prepositions and entity co-occurrences that reflect geospatial configurations. Furthermore, instead of relying solely on syntactic parsing, our method leverages semantic prompting to extract domain-adapted triplets more reliably.

### C. Local-Guided Feature Fusion

To enhance the alignment between scene-level descriptions and object-level semantics in RS retrieval, we introduce the LGFF module (see Fig. 6). This component enables selective integration of holistic context and fine-grained entity features by injecting global guidance into structured local representations.

Let $V_g(T_g) \in \mathbb{R}^d$ be the scene-level embedding extracted from a textual description, and $V_l(T_l) \in \mathbb{R}^{n\times d}$ the object-level features obtained from the ERG, representing $n$ extracted entities. We formulate a cross-level attention mechanism where the scene-level embedding serves as a query to identify and

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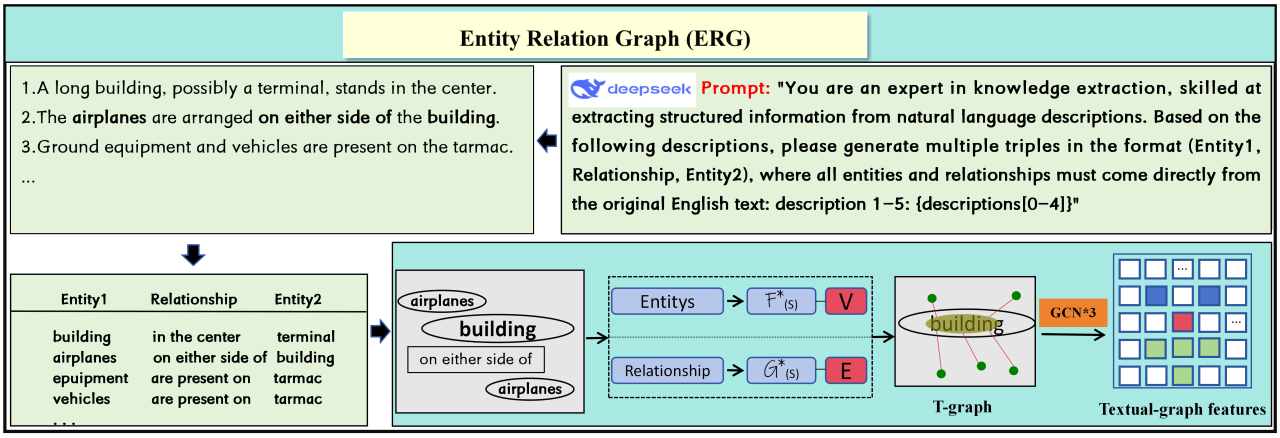



Fig. 5. Architecture of the ERG module. Natural language descriptions are parsed into structured triplets using a large language model prompting. The beddings are constructed and fed into a GCN for structural representation learning.

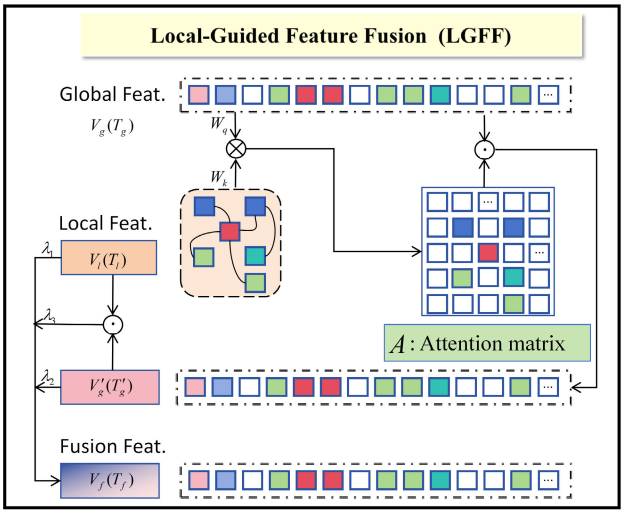



Fig. 6. Architecture of the LGFF module. Scene-level description features attend to structured object-level entities to produce aligned fusion embeddings.

aggregate semantically relevant entity-level nodes. The embeddings are projected into a shared attention

$$Q = V_g\left(T_g\right) W_q$$
$$K = V_l\left(T_l\right) W_k \tag{20}$$

where $W_q, W_k \in \mathbb{R}^{d\times d}$ are the learnable projection matrices. The alignment weights are computed using dot-product attention

$$A = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d}}\right) \in \mathbb{R}^{1\times n}. \tag{21}$$

This attention vector is then used to derive a context-guided semantic vector

$$V_g'\left(T_g'\right) = A \cdot V_l\left(T_l\right) \in \mathbb{R}^d \tag{22}$$

which selectively aggregates object-level features that are most semantically relevant to the overall scene description.

To integrate both the fine-grained object-level semantics and the scene-level contextual information, we employ a residual-like linear fusion mechanism between the local textual features and the scene-guided representation. Formally, the fused feature matrix is computed as follows:

$$V_f\left(T_f\right) = \lambda_1 V_l\left(T_l\right) + \lambda_2 V_g'\left(T_g'\right) \otimes \mathbf{1}_n \tag{23}$$

where $V_l(T_l) \in \mathbb{R}^{n\times d}$ denotes the ERG-derived object-level textual features, $V_g'(T_g') \in \mathbb{R}^{1\times d}$ is the scene-guided global semantic vector, derived from $T_g'$ via a linear projection, $\mathbf{1}_n \in \mathbb{R}^{n\times 1}$ is an all-one vector, used to broadcast $V_g'$ across $n$ textual entity nodes, and $\lambda_1$ and $\lambda_2$ are the scalar weights controlling the relative importance of local and global signals and are learned during training.

This fusion strategy allows each node in the textual graph to retain its unique entity-level semantics from the ERG structure, while being globally informed by the high-level context of the scene. The resulting $V_f(T_f) \in \mathbb{R}^{n\times d}$ serves as a unified textual representation that captures both compositional details and contextual alignment, which is essential for accurate matching with visual object graphs in downstream cross-modal retrieval.

Unlike conventional aggregation methods, LGFF enables the scene description to explicitly guide the fusion of object-level semantics. This design is particularly important in RS scenarios, where textual references often describe spatial relations among objects (e.g., "airplanes on either side of the terminal") that require contextual coherence across scales. By jointly leveraging holistic understanding and localized structure, LGFF bridges the semantic gap between high-level descriptions and grounded object representations, improving retrieval robustness in complex aerial images.

### D. Loss Function and Training Procedure

To achieve consistent cross-modal alignment at both the scene and object levels, we propose HGAC-Loss. This loss consists of two components: a global scene-level triplet loss $L_1$ and a local object-level graph alignment loss $L_2$, which jointly supervise the representations learned from the CMIM interaction and the SOG–ERG structural graphs.

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*1) Scene-Level Triplet Loss $L_1$:* Let $\{(I_i, T_i)\}_{i=1}^{N}$ be a batch of $N$ image–text pairs. We obtain scene-level visual and textual embeddings from CMIM

$$V_i^{(g)} = f_{\text{visual}}(I_i), \quad T_i^{(g)} = f_{\text{text}}(T_i), \quad V_i^{(g)}, T_i^{(g)} \in \mathbb{R}^d. \tag{24}$$

We compute cosine similarity between all visual–text pairs

$$S_{ij} = \cos\left(V_i^{(g)}, T_j^{(g)}\right) = \frac{V_i^{(g)} \cdot T_j^{(g)}}{\|V_i^{(g)}\| \cdot \|T_j^{(g)}\|}, \quad S_{ij} \in [-1, 1]. \tag{25}$$

The triplet loss encourages correct matches ($i = j$) to be more similar than mismatches ($i \neq j$) by a margin $\alpha$

$$L_1 = \sum_{i=1}^{N} \sum_{j \neq i} \max\left(0, S_{ij} - S_{ii} + \alpha\right) \tag{26}$$

where $V_i^{(g)}$ represents the scene-level visual feature of image $I_i$, $T_j^{(g)}$ denotes the scene-level text feature of caption $T_j$, and $\alpha$ is the margin hyperparameter (typically ranging from 0.2 to 0.5).

*2) Object-Level Graph Alignment Loss $L_2$:* This term aligns the object-level structures between the SOG and ERG for each sample. It consists of node-level alignment $L_{\text{node}}$ and structure-level alignment $L_{\text{graph}}$.

*a) Node alignment with object–entity correspondences:* Let $V_i^{(l)} = [v_v^{(i,1)}, \ldots, v_v^{(i,m)}] \in \mathbb{R}^{m \times d}$ be object-level features from the SOG of image $I_i$, and $T_i^{(l)} = [v_t^{(i,1)}, \ldots, v_t^{(i,n)}] \in \mathbb{R}^{n \times d}$ from the ERG of text $T_i$. We compute the pairwise cosine similarity

$$S_{kl}^{(i)} = \cos\left(v_v^{(i,k)}, v_t^{(i,l)}\right), \quad k \in [1, m], \, l \in [1, n]. \tag{27}$$

Using the Hungarian algorithm, we find the maximum-weight bipartite matching

$$M_i^* = \arg \max_{M_i \subseteq [m] \times [n]} \sum_{(k,l) \in M_i} S_{kl}^{(i)}. \tag{28}$$

Then, the node loss is

$$L_{\text{node}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{(k,l) \in M_i^*} S_{kl}^{(i)}. \tag{29}$$

*b) Structural alignment with spatial relations:* To enhance the topological consistency across modalities, we align the adjacency structures of the SOG and the ERG. Unlike prior works that rely solely on node correspondence, our method explicitly considers spatial relations types for fine-grained structural alignment.

*Visual Graph Adjacency With Spatial Relation Types:* For the SOG, we go beyond raw distances and model discrete spatial relations between object pairs. Given two detected objects $o_k$ and $o_{k'}$ with normalized centroids $P_k^{(i)}$ and $P_{k'}^{(i)}$, we compute a spatial relation label $s_{kk'}^{(i)} \in \mathcal{S}$ from a predefined set (e.g., *left of*, *right of*, *above*, *below*, *inside*, and *adjacent*).

We encode spatial relations via a trainable embedding

$$R_{kk'}^{(i)} = E_s\left(s_{kk'}^{(i)}\right) \in \mathbb{R}^d \tag{30}$$

where $E_s(\cdot)$ maps discrete spatial relation types to $d$-dimensional vectors. Then, the enriched adjacency is defined

$$A_v^{(i)}(k, k') = \cos\left(v^{(i,k)} + R_{kk'}^{(i)}, \, v^{(i,k')}\right). \tag{31}$$

This design allows each visual edge to carry spatial semantics, enabling more expressive graph propagation and better structural alignment.

*Textual Graph Adjacency With Relation Types:* For the ERG, we similarly construct relation-aware edges by encoding both textual entities and their semantic relations

$$A_t^{(i)}(l, l') = \cos\left(E\left(v_t^{(i,l)}\right) + E\left(r_t^{(i,l,l')}\right), \, E\left(v_t^{(i,l')}\right)\right) \tag{32}$$

where $E(\cdot)$ is a CLIP-based encoder for both entity and relation phrases. If no explicit relation exists between $v_l$ and $v_{l'}$, we set $E(r_t^{(i,l,l')}) = \mathbf{0}$.

*Structural Alignment Loss:* We normalize both adjacency matrices using rowwise $\ell_2$ normalization

$$\bar{A}_v^{(i)} = \text{Norm}\left(A_v^{(i)}\right), \quad \bar{A}_t^{(i)} = \text{Norm}\left(A_t^{(i)}\right). \tag{33}$$

Then, compute the Frobenius norm of their difference

$$L_{\text{graph}} = \frac{1}{N} \sum_{i=1}^{N} \left\| \bar{A}_v^{(i)} - \bar{A}_t^{(i)} \right\|_F^2. \tag{34}$$

This formulation encourages modality-agnostic topological alignment, accounting for both spatial relational.

*c) Overall training objective:* The final training loss integrates scene-level and object-level graph consistency

$$L_{\text{HGAC}} = L_1 + \lambda\left(L_{\text{node}} + L_{\text{graph}}\right). \tag{35}$$

Here, $\lambda$ is a tunable weight balancing global and local signals. This hierarchy enables GraVLM to capture both scene-level semantics and structured spatial arrangements across modalities.

*3) Training Procedure:* The training procedure of the proposed GraVLM is outlined in Algorithm 1. During the training process, paired RS image–text datasets $\{(I_i, T_i)\}_{i=1}^{N}$ are used. For the image feature representation, we first extract features using a pretrained vision encoder (e.g., CLIP ViT-B/16) to obtain the initial image embeddings. Then, GraVLM processes these features through the SOG module to capture object-level semantics, followed by the ERG module to capture relational semantics from the text description. These embeddings are further refined by the LGFF module, which integrates the object-level and scene-level features, yielding a refined textual feature representation. Finally, the HGAC-Loss is applied to jointly optimize the alignment between object–entity correspondences and spatial relations across modalities.

The end-to-end training procedure optimizes both image and text embeddings while ensuring the preservation of fine-grained semantic alignment between the two modalities.

## IV. Experiments

### A. Datasets and Experimental Setup

*1) Datasets:* We evaluate our model on two representative RS datasets: RS5M and RSITMD. RS5M is a large-scale dataset containing over 5 million high-resolution RS images paired. It serves as a pretraining corpus for VLMs in the RS domain. RSITMD provides manually annotated image–text pairs with object-level correspondences and relational semantics, enabling fine-grained evaluation of cross-modal retrieval performance.

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**Algorithm 1** Training Procedure of GraVLM

**Input:** Image–text pairs $\{(I_i, T_i)\}_{i=1}^{N}$
**Input:** Margin $\alpha$ and weight $\lambda$
**Output:** Optimized model parameters $\Theta$

**for** each training iteration **do**
  Sample a mini-batch $\mathcal{B} = \{(I_i, T_i)\}_{i=1}^{B}$
  **for** each $(I_i, T_i) \in \mathcal{B}$ **do**
    Encode $I_i$ and $T_i$ into token embeddings $v_0^{(i)} \in \mathbb{R}^{N_v \times d}$ and $t_0^{(i)} \in \mathbb{R}^{N_t \times d}$
    Apply CMIM to obtain scene-level embeddings $V_i^{(g)}$ and $T_i^{(g)}$
    Detect objects in $I_i$, construct SOG, and encode object-level features $V_i^{(l)}$ via SOG-GCN
    Extract entity–relation triplets from $T_i$, construct ERG, and encode entity-level features $T_i^{(l)}$ via ERG-GCN
    Apply LGFF to fuse $T_i^{(g)}$ and $T_i^{(l)}$, obtaining fused textual features $T_i^{(f)}$
    Compute node alignment loss $L_{\text{node}}^{(i)}$ (Hungarian matching)
    Compute structure-level alignment loss $L_{\text{graph}}^{(i)}$
  **end for**
  Compute scene-level triplet loss $L_1$ over $\mathcal{B}$
  Compute object-level loss $L_2 = \frac{1}{B}\sum_{i=1}^{B}(L_{\text{node}}^{(i)} + L_{\text{graph}}^{(i)})$
  Update $\Theta$ by minimizing $L_{\text{HGAC}} = L_1 + \lambda L_2$
**end for**

*2) Methods Included for Comparison:* We compare GraVLM with representative RS text–image retrieval methods from two groups: 1) methods without VLMs and 2) methods with VLMs.

1) *Methods Without VLMs:* This group covers conventional cross-modal retrieval models that are trained from scratch or rely on task-specific encoders, including CAMP [28], LW-MCR [29], AMFMN [11], KACM [12], GaLR [17], SCAN [30], PIR [31], FAAMI [13], MTGFE [18], MSCMAT [15], IERR [32], QDMVR [33], VGSM [19], SIPN [16], and FSSN [22].
2) *Methods With VLMs:* This group includes VLM-based RS retrieval methods that leverage large-scale image–text pretraining and domain adaptation, including CLIP-Baseline, PETL [25], SkyScript [34], GRAFT [35], RemoteCLIP [4], and GeoRSCLIP [24]. For VLM-based baselines, we further report both the zero-shot results (without task-specific fine-tuning) and the fine-tuned results when applicable (denoted by "+FT"), including CLIP + FT and GeoRSCLIP + FT [24].

We report results for both our base model (GraVLM) and its fine-tuned variant (GraVLM+FT) under zero-shot and few-shot evaluation protocols.

*3) Evaluation Metrics:* We adopt two primary tasks: image-to-text (I2T) retrieval and text-to-image (T2I) retrieval and use *Recall@K* and *mR* as evaluation metrics. The calculation of *Recall@K* is defined as follows:

$$\text{R@}K = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}\left[\text{rank}\,(y_i) \le K\right] \tag{36}$$

where $N$ denotes the total number of queries in the test set, $y_i$ is the ground-truth match for the $i$th query, $\text{rank}(y_i)$ is its rank in the retrieved result list, and $\mathbf{1}[\cdot]$ is the indicator function that returns 1 if the condition is true, and 0 otherwise. The final metric *mR* is computed as the average of *Recall@K* (*R@K*) for $K \in \{1, 5, 10\}$ over both I2T and T2I tasks.

*4) Implementation Details:* GraVLM is instantiated on top of CLIP-style vision and text encoders, upon which we build lightweight graph modules for object-level structural modeling. Unless otherwise stated, all experiments use CLIP ViT-B/16 as the vision backbone and the standard CLIP text Transformer as the text backbone. On top of the scene-level embeddings produced by these backbones, we employ two graph encoders (SOG and ERG) as object-level embeddings. This design isolates the proposed hierarchical graph modeling from the backbone capacity, while keeping the additional parameter overhead modest.

TABLE I
ENCODER SPECIFICATIONS AND PARAMETER COUNTS USED IN GRAVLM

| Encoder | Architecture | Layers | Hidden Dim | Params |
|---|---|---|---|---|
| Vision Encoder | CLIP ViT-B/16 | 12 | 768 | 86M |
| Text Encoder | CLIP Text Transformer | 12 | 512 | 63M |
| SOG Encoder | 3-layer GCN | 3 | 256 | ∼2.1M |
| ERG Encoder | 3-layer GCN | 3 | 256 | ∼2.3M |

Table I reports the architectural specifications and parameter counts of each component. The two CLIP backbones account for the majority of model parameters (86M for the vision encoder and 63M for the text encoder), whereas the proposed graph encoders introduce only a small number of additional parameters (approximately 2.1M/2.3M for SOG/ERG, respectively). As a result, the overall model size remains dominated by the pretrained encoders, and the graph-based structural reasoning can be integrated with negligible parameter increase.

All models are trained on a single NVIDIA A100 GPU with a batch size of 128. Other optimization hyperparameters follow standard practice and are reported in the experimental setup for reproducibility.

## B. Experimental Results and Analysis

Table II reports a comprehensive comparison with state-of-the-art RS cross-modal retrieval methods, including methods without VLMs, VLM-based methods, and fine-tuned VLM (VLM+FT). GraVLM achieves superior and consistent improvements across evaluation settings.

*1) Zero-Shot Setting:* The zero-shot setting is particularly important for RS retrieval, where collecting target-domain annotations can be expensive. To evaluate cross-domain generalization, we pretrain VLM-based methods on RS5M and directly test on RSITMD without using any target-domain supervision. As reported in Table II, GraVLM achieves the best overall zero-shot performance among all VLM-based methods. Specifically, for I2T retrieval, GraVLM obtains Recall@1/5/10 scores of 33.15%, 63.82%, and 69.21%, respectively. For T2I retrieval, it achieves 37.25%, 64.14%,

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TABLE II
COMPARISON OF STATE-OF-THE-ART METHODS ON THE RSITMD BENCHMARK FOR CROSS-MODAL RETRIEVAL

| Train→Test | Method | Year | Paradigm | I2T (Recall@K) | | | T2I (Recall@K) | | | mR |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 | |
| T*0.8→T*0.2 | CAMP | 2019 | w/o VLM | 3.87 | 9.92 | 15.14 | 2.74 | 11.63 | 21.37 | 10.78 |
| T*0.8→T*0.2 | LW-MCR | 2021 | w/o VLM | 4.33 | 13.25 | 19.18 | 3.84 | 14.96 | 25.73 | 13.55 |
| T*0.8→T*0.2 | AMFMN | 2021 | w/o VLM | 4.98 | 14.17 | 20.75 | 4.92 | 17.83 | 28.33 | 15.51 |
| T*0.8→T*0.2 | KACM | 2022 | w/o VLM | 6.42 | 18.69 | 36.58 | 7.68 | 22.31 | 36.62 | 21.38 |
| T*0.8→T*0.2 | GaLR | 2022 | w/o VLM | 11.15 | 36.68 | 51.68 | 14.82 | 31.64 | 42.48 | 31.41 |
| T*0.8→T*0.2 | SCAN | 2023 | w/o VLM | 4.91 | 12.43 | 19.62 | 4.58 | 16.21 | 26.43 | 14.41 |
| T*0.8→T*0.2 | PIR | 2023 | w/o VLM | 18.14 | 41.15 | 52.88 | 12.17 | 41.68 | 63.41 | 38.24 |
| T*0.8→T*0.2 | FAAMI | 2023 | w/o VLM | 16.15 | 35.62 | 48.89 | 12.96 | 42.39 | 59.96 | 35.99 |
| T*0.8→T*0.2 | MTGFE | 2023 | w/o VLM | 16.33 | 48.15 | 66.85 | 17.18 | 40.27 | 53.48 | 40.38 |
| T*0.8→T*0.2 | MSCMAT | 2024 | w/o VLM | 22.35 | 42.92 | 55.75 | 15.18 | 47.35 | 64.73 | 41.38 |
| T*0.8→T*0.2 | IERR | 2024 | w/o VLM | 27.66 | 48.89 | 61.06 | 24.43 | 56.59 | 72.66 | 48.55 |
| T*0.8→T*0.2 | QDMVR | 2024 | w/o VLM | 19.22 | 45.41 | 58.89 | 13.98 | 47.63 | 66.21 | 41.89 |
| T*0.8→T*0.2 | VGSM | 2024 | w/o VLM | 14.16 | 34.96 | 50.66 | 13.23 | 42.57 | 63.43 | 36.51 |
| T*0.8→T*0.2 | SIPN | 2025 | w/o VLM | 23.01 | 46.02 | 57.96 | 17.04 | 49.87 | 66.95 | 43.47 |
| T*0.8→T*0.2 | FSSN | 2025 | w/o VLM | 30.53 | 52.21 | 65.49 | 25.71 | 58.32 | **75.40** | 51.28 |
| RS5M→T*1 | CLIP-Baseline | — | w/ VLM | 9.11 | 20.42 | 31.70 | 8.46 | 27.18 | 39.98 | 22.81 |
| RS5M→T*1 | PETL | 2023 | w/ VLM | 24.16 | 47.12 | 61.28 | 20.41 | 50.53 | 68.54 | 45.33 |
| RS5M→T*1 | SkyScript | 2024 | w/ VLM | 12.05 | 35.74 | 53.68 | 13.81 | 33.14 | 40.48 | 31.48 |
| RS5M→T*1 | GRAFT | 2024 | w/ VLM | 25.25 | 44.68 | 57.89 | 24.92 | 53.39 | 61.47 | 44.61 |
| RS5M→T*1 | RemoteCLIP | 2024 | w/ VLM | 27.88 | 50.66 | 65.71 | 22.17 | 56.46 | 73.41 | 49.38 |
| RS5M→T*1 | GeoRSCLIP | 2024 | w/ VLM | 30.09 | 51.55 | 63.27 | 23.54 | 57.52 | 74.60 | 50.10 |
| RS5M→T*1 | **GraVLM** | **ours** | w/ VLM | **33.15** | **63.82** | **69.21** | **35.21** | **64.14** | 70.02 | **55.93** |
| RS5M→T*0.5 | CLIP+FT | — | VLM+FT | 13.52 | 39.55 | 50.61 | 15.25 | 39.17 | 52.93 | 35.67 |
| RS5M→T*0.5 | GeoRSCLIP+FT | 2024 | VLM+FT | 32.30 | 53.32 | 67.92 | 25.04 | 57.88 | 74.38 | 51.81 |
| RS5M→T*0.5 | **GraVLM+FT** | **ours** | VLM+FT | **39.32** | **69.78** | **74.52** | **41.70** | **67.25** | **74.63** | **61.20** |

and 70.02% at Recall@1/5/10, respectively, leading to an overall mean Recall (*mR*) of 55.93%.

We attribute these gains to GraVLM's hierarchical design that couples scene-level cross-modal interaction with object-level graph structural modeling, which encourages more discriminative and structure-aware alignment between RS images and textual descriptions under domain shift.

*2) Few-Shot Setting:* To evaluate THE model adaptability under limited supervision, we conduct few-shot experiments by fine-tuning VLM-based methods on only 50% of the RSITMD training data. As shown in Table II, GraVLM+FT achieves the best overall performance among all compared methods in the few-shot setting.

Specifically, for I2T retrieval, GraVLM+FT attains Recall@1/5/10 scores of 39.32%, 69.78%, and 74.52%, respectively. For T2I retrieval, it achieves 41.70%, 67.25%, and 74.63% at Recall@1/5/10, resulting in an overall mR of 61.20%. Compared with the strongest fine-tuned baseline, GeoRSCLIP+FT (51.81% mR), GraVLM+FT yields a substantial improvement under the few-shot setting.

These results indicate that the proposed hierarchical graph-aligned modeling remains effective even when only limited target-domain annotations are available. By explicitly modeling object-centric structures and cross-modal relations through SOG, ERG, and GraVLM exhibits improved robustness in few-shot RS cross-modal retrieval.

### C. Ablation Study

The above experiments have demonstrated the overall effectiveness of GraVLM on RS cross-modal retrieval benchmarks. To further quantify the contribution of each key component and better understand their roles in the proposed hierarchical graph-aligned framework, we conduct an ablation study with a series of controlled model variants. Specifically, we define six variants (P0–P5) by progressively integrating core components into the CLIP baseline and evaluate the performance impact of each module.

1) *P0:* CLIP baseline without any cross-modal interaction or structure modeling.
2) *P1:* P0 + CMIM.
3) *P2:* P1 + SOG for visual structural reasoning.
4) *P3:* P2 + ERG for textual compositional semantics.
5) *P4:* P3 + LGFF.
6) *P5:* P4 + HGAC-Loss.

To investigate the contribution of each key component in GraVLM, we conduct an incremental ablation study on RSITMD. The results are reported in Table III.

The baseline model P0 exhibits limited retrieval performance (mR = 22.81%), as it relies solely on independent image and text encoders without any explicit cross-modal interaction or structural modeling. After introducing the CMIM in P1, the mR increases to 34.44%, indicating that explicit scene-level cross-modal interaction helps mitigate modality gaps even without object-level structures for RS cross-modal retrieval.

By further incorporating the SOG in P2, the model gains the ability to encode object-level spatial structures from RS images, leading to a notable improvement in mR to 44.49%. When the ERG is added in P3, the model additionally captures compositional semantics and entity relations from textual descriptions, resulting in a further performance gain (mR = 52.75%). These results suggest a complementary effect

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TABLE III
INCREMENTAL ABLATION STUDY ON THE RSITMD BENCHMARK

| ID | CMIM | SOG | ERG | LGFF | HGAC | I2T (Recall@K) | | | T2I (Recall@K) | | | mR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | R@1 | R@5 | R@10 | R@1 | R@5 | R@10 | |
| P0 | | | | | | 9.11 | 20.42 | 31.70 | 8.46 | 27.18 | 39.98 | 22.81 |
| P1 | ✓ | | | | | 17.88 | 38.60 | 52.47 | 14.92 | 34.87 | 50.91 | 34.44 |
| P2 | ✓ | ✓ | | | | 25.16 | 49.85 | 63.33 | 22.37 | 45.73 | 61.52 | 44.49 |
| P3 | ✓ | ✓ | ✓ | | | 31.03 | 58.92 | 71.04 | 28.85 | 54.60 | 70.07 | 52.75 |
| P4 | ✓ | ✓ | ✓ | ✓ | | 36.42 | 65.10 | 74.89 | 34.91 | 62.14 | 76.11 | 58.93 |
| P5 | ✓ | ✓ | ✓ | ✓ | ✓ | **39.32** | **69.78** | **74.52** | **41.70** | **67.25** | **74.63** | **61.20** |

TABLE IV
MODEL COMPLEXITY COMPARISON

| Model | Backbone | Params | FLOPs |
|---|---|---|---|
| RemoteCLIP (2024) | ViT-B/16 + CLIP-T | ∼149M | ∼19.8G |
| GeoRSCLIP (2024) | ViT-B/16 + CLIP-T | ∼150M | ∼20.0G |
| GraVLM (Ours) | ViT-B/16 + CLIP-T | ∼153M | ∼20.0G |

TABLE V
EFFECT OF DIFFERENT CLIP VISION ENCODERS ON GRAVLM

| Vision Encoder | Params | FLOPs | mR (%) |
|---|---|---|---|
| CLIP ViT-B/16 | ∼153M | ∼20G | 55.9 |
| CLIP ViT-L/14 | ∼304M | ∼60G | 56.2 |
| CLIP ViT-H/14 | ∼632M | ∼180G | 56.9 |

between SOG and ERG, where visual object structures and textual relational semantics jointly facilitate fine-grained cross-modal correspondence.

In P4, the LGFF module is introduced to integrate scene-level and object-level representations, which further strengthens cross-modal alignment and improves the mR to 58.93%. Finally, P5 incorporates the proposed HGAC-Loss, which explicitly enforces consistency at both the entity correspondence and relational structure levels. This hierarchical supervision yields the best overall performance, with the mR reaching 61.20%.

Overall, the incremental ablation results confirm that each component of GraVLM contributes positively to retrieval performance. In particular, object-centric graph modeling (SOG and ERG) and graph-aligned contrastive supervision play a key role in enabling accurate and robust RS cross-modal retrieval.

### D. Model Complexity Analysis

This section analyzes the computational overhead of GraVLM from two perspectives: 1) model size (Params) and 2) computational cost (FLOPs).

*1) Comparison With Existing Methods:* All results are reported under the zero-shot setting on RSITMD. Table IV lists the comparison of GraVLM with representative RS-oriented VLM retrieval methods under the same CLIP ViT-B/16 backbone. Under this configuration, GraVLM contains approximately 153M parameters in total, where the structural components (SOG, ERG, and LGFF) introduce around 3M additional parameters (less than 2% of the total). The computational cost is about 20 GFLOPs per forward pass at an input resolution of $224 \times 224$, which is dominated by the backbone vision and text encoders. Despite incorporating hierarchical graph-based modeling, GraVLM maintains a complexity level comparable to RemoteCLIP and GeoRSCLIP, indicating that its performance gains are not obtained at the cost of excessive model complexity.

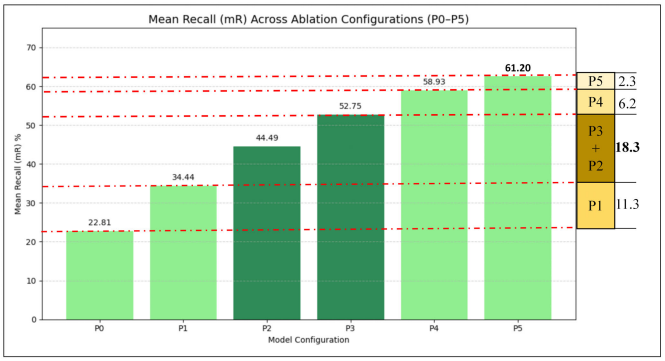



Fig. 7. Mean Recall (mR) comparison under different ablation configurations (P0–P5). Local representation modules (P2 and P3) bring substantial improvement, while LGFF and HGAC-Loss offer additional gains.

*2) Impact of Different Encoder Choices:* To analyze the performance–efficiency tradeoff under different backbone capacities, we evaluate GraVLM with several CLIP vision encoders while keeping the proposed graph modules fixed. Table V reports the mR performance together with the corresponding Params and FLOPs.

As shown in Table V, larger CLIP backbones improve the absolute retrieval accuracy but incur substantially higher parameter and computational costs. Notably, increasing the backbone capacity leads to incremental performance gains for GraVLM. Meanwhile, the proposed graph-aligned modeling remains effective across different backbone scales, suggesting that its contribution is complementary to, rather than dominated by, backbone scaling. Considering both performance and efficiency, we adopt CLIP ViT-B/16 as the default backbone in our experiments.

### E. Visualization Analysis

*1) Ablation Visualization and Module Contribution Analysis:* To further investigate the impact of each component in GraVLM, we visualize the mean Recall (mR) under different

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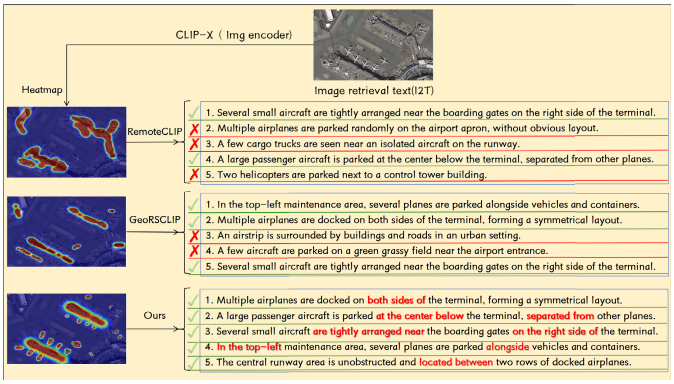



Fig. 8. Qualitative comparison of attention heatmaps and Top-5 retrieved sentences for the I2T task. While RemoteCLIP and GeoRSCLIP exhibit scattered or imprecise attention, our GraVLM model focuses sharply on aircraft clusters and terminal structures. The retrieved captions from GraVLM more accurately reflect spatial relations, validating the effectiveness of its object-level graph modeling and hierarchical alignment.

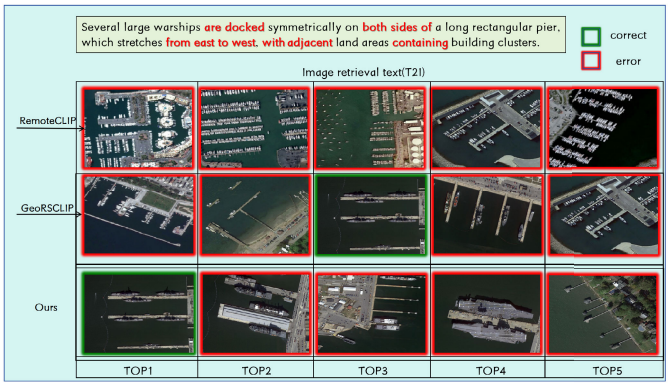



Fig. 9. Qualitative comparison of Top-5 T2I retrieval results. Our GraVLM better captures the object-level layout and directional relations described in the query (highlighted in red), while prior methods often retrieve semantically or structurally inconsistent scenes.

configurations (P0–P5) based on Table III, as shown in Fig. 7. Specifically, integrating the SOG in P2 enables the model to capture spatial structures among objects, resulting in a significant mR increase from 34.44% to 44.49%. Subsequently, incorporating the ERG in P3 allows the model to embed structured textual semantics through entity relations, further boosting mR to 52.75%. These two object-level modules cumulatively contribute an 18.31% improvement, confirming the effectiveness of local structural modeling for fine-grained vision–language alignment.

Furthermore, the introduction of the LGFF in P4 facilitates explicit interaction between global scene-level and local object-level representations, improving attention to salient regions and increasing mR to 58.93%. Finally, the HGAC-Loss introduced in P5 strengthens both node and structure consistency between SOG and ERG. This results in the best performance across all metrics, with I2T and T2I Recall@1 reaching 39.32% and 41.70%, and mR achieving 61.20%. These results demonstrate that each module in GraVLM contributes progressively, and that hierarchical graph-aligned modeling plays a critical role in enhancing cross-modal retrieval robustness in RS.

*2) Visual Assessment of I2T Retrieval:* To qualitatively assess the effectiveness of our structure-aware alignment framework, we visualize the attention distributions and retrieval outputs for a representative I2T query. Fig. 8 shows the comparison of RemoteCLIP, GeoRSCLIP, and our GraVLM on an RS image containing multiple aircraft and terminal structures.

RemoteCLIP exhibits diffuse attention, failing to concentrate on semantically meaningful regions. GeoRSCLIP provides partial focus on relevant targets but lacks spatial precision. In contrast, GraVLM, guided by its SOG, generates more compact and accurate attention over salient objects such as aircraft clusters and airport terminals.

*3) Visual Assessment of T2I Retrieval:* To further demonstrate the superiority of GraVLM in cross-modal alignment, we present a qualitative comparison of Top-5 T2I retrieval results in Fig. 9. The input query emphasizes complex spatial semantics, such as "warships docked symmetrically on both sides of a rectangular pier" and "from east to west," requiring fine-grained object-level and spatial relationship understanding.

As shown, RemoteCLIP and GeoRSCLIP fail to retrieve relevant scenes, predominantly returning dense marina scenes or irrelevant dock arrangements. Their Top-5 results exhibit semantic drift from the query, primarily due to a lack of structural reasoning and insufficient alignment at the object-relation level.

In contrast, GraVLM successfully retrieves images that match both the visual entities (warships and rectangular pier) and their spatial configurations (symmetry and directionality). The Top-1 and Top-3 retrieved images precisely reflect the described layout, highlighting the model's capability to capture structured semantics. This performance gain is attributed to the joint modeling of SOG and ERG, as well as the HGAC, which enforces consistency at both node and structural levels.

These results clearly validate the effectiveness of our proposed framework in handling complex spatial layouts and structured semantics, which are prevalent in RS images but difficult to model using global-only vision–language paradigms.

### F. Failure Case Analysis

To gain deeper insight into the behavior of GraVLM in RS cross-modal retrieval, we analyze representative failure cases on the RSITMD testing set. We examine failure patterns from both the vision side and text side of the proposed graph-aligned model.

*1) Vision-Side Failures:* From the vision-side perspective, most retrieval failures are related to ambiguities in object grounding and spatial structure modeling. Two dominant error patterns are observed, as illustrated in Fig. 10. The first type is *object–number mismatch* (approximately 21% of errors), which typically occurs when the caption specifies a precise count of objects—such as airplanes or vehicles—but the visual object extraction module fails to identify small or overlapping instances. Since the SOG is constructed directly from detected objects, missing nodes lead to an incomplete structural representation and consequently incorrect cross-modal alignment.

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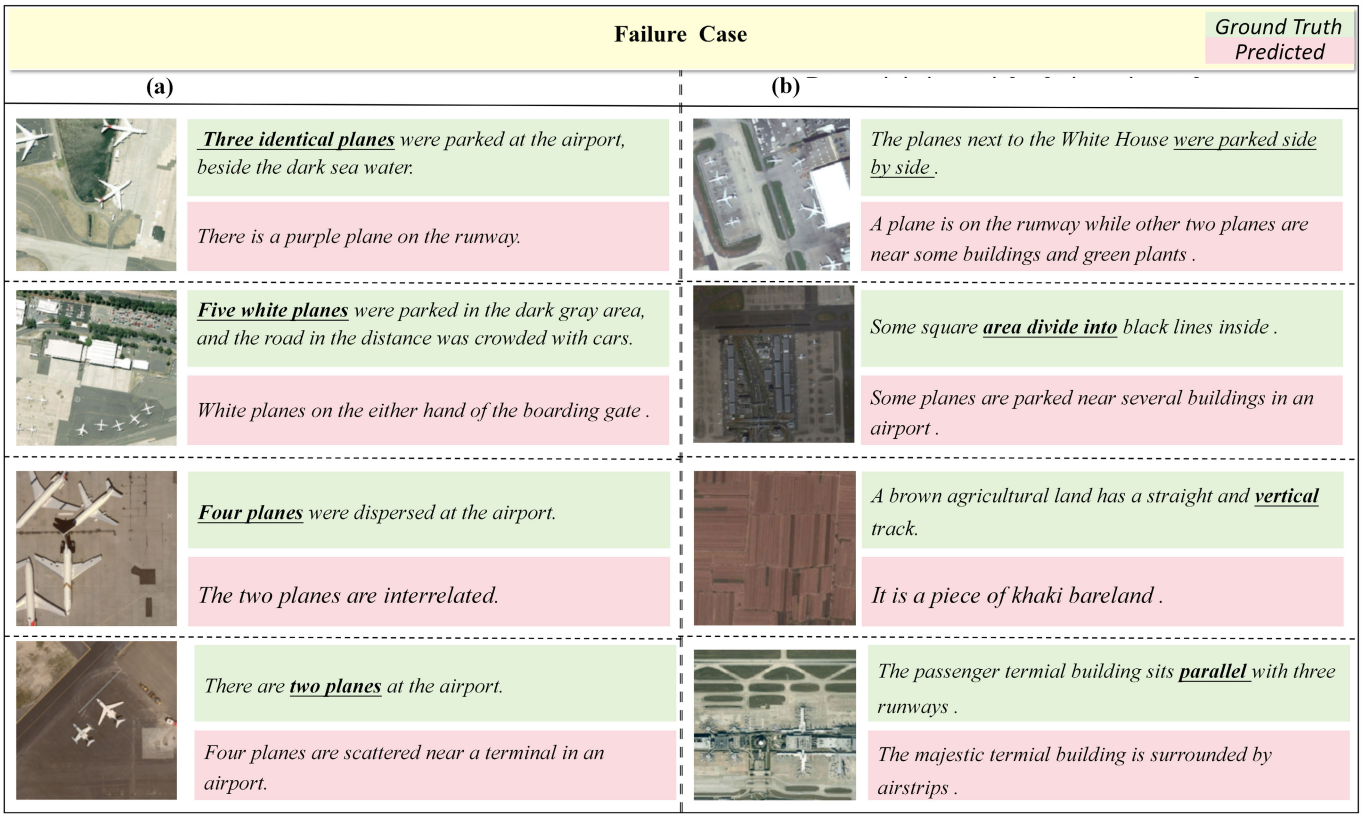



Fig. 10. Representative failure cases of GraVLM, including (a) object–number mismatches and (b) deterministic spatial–relation mismatches. Green text indicates ground truth, while red text denotes incorrect predictions.

TABLE VI
HUMAN-VERIFIED ACCURACY OF TEXT-SIDE TRIPLE EXTRACTION METHODS

| Method | Accuracy (%) |
|---|---|
| Stanford CoreNLP Parser | 72 |
| Qwen2.5 (7B) | 94 |
| DeepSeek-V3 | 95 |

The second category involves *deterministic spatial–relation mismatch* (around 9% of errors), including relations such as "parallel," "vertical," or "aligned with." These cases arise when the ERG fails to capture geometric relations accurately, causing misalignment despite correct object recognition. Additional failures are observed in visually cluttered or highly heterogeneous scenes, where LGFF may fuse noisy local features. Overall, the errors mainly stem from imperfect object detection, incomplete relation extraction, and complex remote sensing environments, highlighting opportunities for stronger geometry-aware relation modeling and more robust multiscale feature integration.

*2) Text-Side Failures:* In addition to vision-side errors, we further analyze failure cases originating from the text-side entity and relation extraction module. Since the ERG is constructed from structured entity–relation triples, unreliable triples may introduce noise into graph-based cross-modal alignment if not properly handled.

To quantitatively evaluate the reliability of text-side extraction, we benchmark three representative methods: 1) a traditional dependency-based NLP parser (Stanford CoreNLP); 2) Qwen2.5 (7B); and 3) DeepSeek-V3. We randomly sample 400 captions from the RSITMD training set and manually verify whether the extracted triples correctly reflect the entities and their pairwise relations described in the original text. The human-verified accuracy results are reported in Table VI. Among all evaluated methods, DeepSeek-V3 achieves the highest accuracy and is, therefore, adopted in GraVLM.

Despite the high overall accuracy of DeepSeek-V3, a small number of extraction errors are still observed during manual inspection. These errors can be categorized into three types: 1) missing entities or relations in long or syntactically complex descriptions; 2) incorrect or ambiguous spatial relations, especially for fine-grained geometric relations (e.g., parallel, across, or surrounding); and 3) hallucinated entities that cannot be grounded in the original caption.

Table VII summarizes several representative failure cases of text-side triple extraction. It is worth noting that, on the one hand, the overall accuracy of LLM-based triple extraction is relatively high; on the other hand, each image in the RSITMD dataset is typically associated with multiple textual descriptions, and the extracted entities and relations across different captions exhibit a certain degree of redundancy and complementarity. This property improves the robustness of the resulting structural semantic representation at the image level.

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TABLE VII
REPRESENTATIVE FAILURE CASES OF TEXT-SIDE TRIPLE EXTRACTION. INCORRECT ELEMENTS ARE HIGHLIGHTED IN RED

| Caption | Extracted triples | Error type |
|---|---|---|
| "two runways intersecting near the terminal building" | (runways, intersecting, terminal) | Incorrect relation |
| "several ships docked along the river beside the industrial area" | (ships, beside, industrial area) | Incorrect relation |
| "residential buildings and warehouses separated by a main road" | (buildings, separated by, road) | Incorrect entity |
| "an airport with aircraft parked around multiple terminals" | (aircraft, around, terminals); (aircraft, parked at, —) | Missing relation |

Based on these observations, GraVLM adopts a conservative postprocessing strategy.

1) Before constructing the ERG, semantic triples with missing entities or relations are directly removed, rather than being corrected or heuristically completed. This strategy helps to prevent potentially unreliable relational edges from being introduced into the graph, thereby reducing noise in cross-modal structural alignment.
2) When all extracted relations from a given caption are filtered out, we fall back to constructing a node-only ERG, which preserves entity-level semantics while avoiding the introduction of noisy or misleading relational connections.

## V. CONCLUSION

In this work, we proposed GraVLM, a novel hierarchical graph-aligned vision–language framework tailored for cross-modal retrieval in RS. Unlike previous methods that rely primarily on the global semantic alignment, GraVLM explicitly models spatial object configurations and compositional semantics through two complementary structures: the SOG for visual domain modeling and the ERG for textual structure representation. To bridge the modality gap, we introduce an HGAC-Loss that aligns entities and relations across modalities at both local and structural levels. By jointly leveraging global scene features and graph-enhanced local representations, GraVLM achieves fine-grained semantic understanding and robust alignment under complex object layouts and sparse supervision. Experimental results on RS5M and RSITMD demonstrate consistent improvements over existing supervised and pretrained baselines in both zero-shot and few-shot settings.

This study highlights the importance of structure-aware modeling for vision–language understanding in RS. In future work, we plan to extend GraVLM to region-level retrieval, multiturn reasoning, and integration with temporal satellite images for broader geospatial applications.

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